

# PortIN — Intelligent Maritime Freight Intelligence Platform

Official SIH 2026 Problem Statement 26006 | SAIL & Ministry of Steel | Generated: 2026-09-03 14:20:36

## 1. Executive Summary

Official executive chartering evaluation. Recommends 3-voyage Panamax Short-Term Contract into Paradip CQ-1, mitigating seasonal freight inflation.

## 2. Operational Chartering Parameters

| Parameter            | Evaluated Value                         | Source Provenance                |
|----------------------|-----------------------------------------|----------------------------------|
| Cargo Commodity      | Coking Coal                             | SAIL Blast Furnace Specification |
| Parcel Size          | 70,000 MT                               | Procurement Requirement          |
| Trade Lane           | Australia -> Paradip                    | Major Maritime Bulk Corridor     |
| Recommended Vessel   | Panamax                                 | Physical Berth Optimization      |
| Contract Strategy    | Short-Term Multi-Voyage (3 Voyages COA) | Volatility Mitigation Model      |
| Market Signal        | BOOK NOW                                | HistGradientBoosting Model       |
| Optimal Entry Window | Next 7-14 Days                          | Forward Freight Curve Analysis   |
| Estimated Total Cost | \$1,825,000.00 USD                      | Aggregated Chartering Model      |
| Operational Risk     | 24.5 / 100 (LOW)                        | Multi-Factor Risk Engine         |

## 3. PortIN Decision Twin Strategy Evaluation

The Decision Twin ran 36 stochastic perturbations simulating variations across weather, berth queues, and bunker rates. Plan A is recommended as the optimum balance of volume savings and draft safety.

| Plan                  | Strategy                     | Vessel             | Rate (\$/MT) | Est. Idle (Days) | Risk            |
|-----------------------|------------------------------|--------------------|--------------|------------------|-----------------|
| Plan A (Best Overall) | Short-Term Multi-Voyage (3x) | Panamax            | \$14.15      | 5.4 days         | 24.5 (Low)      |
| Plan B (Lowest Risk)  | Spot with Guaranteed Berth   | Supramax           | \$15.95      | 3.3 days         | 14.0 (Very Low) |
| Plan C (Lowest Cost)  | Long-Term Strategic COA      | Capesize / Panamax | \$13.05      | 8.4 days         | 52.0 (Medium)   |

## 4. Data Provenance & Trust Layer

NOTICE: Port constraints and berth physical parameters are based on officially gazetted Port Trust publications. Freight curves are generated via statistical Gradient Boosting models trained on reproducible seeded benchmarks (SIMULATED SIH DEMO DATA). Marine weather data integrates Open-Meteo services for situational awareness. This report is for strategic logistics planning and not for operational vessel navigation or contractual commitment.